

Reliable UDP Communication in Mininet

Assignment Report

Name	Supratim Gogoi
Roll Number	CS-BTC24-17
Timeout	1.2 seconds
Topology	h1 → s1 → h2

Objective

To implement Reliable UDP communication using the Stop-and-Wait protocol in Mininet with acknowledgements, timeout, retransmissions and duplicate detection.

Implementation

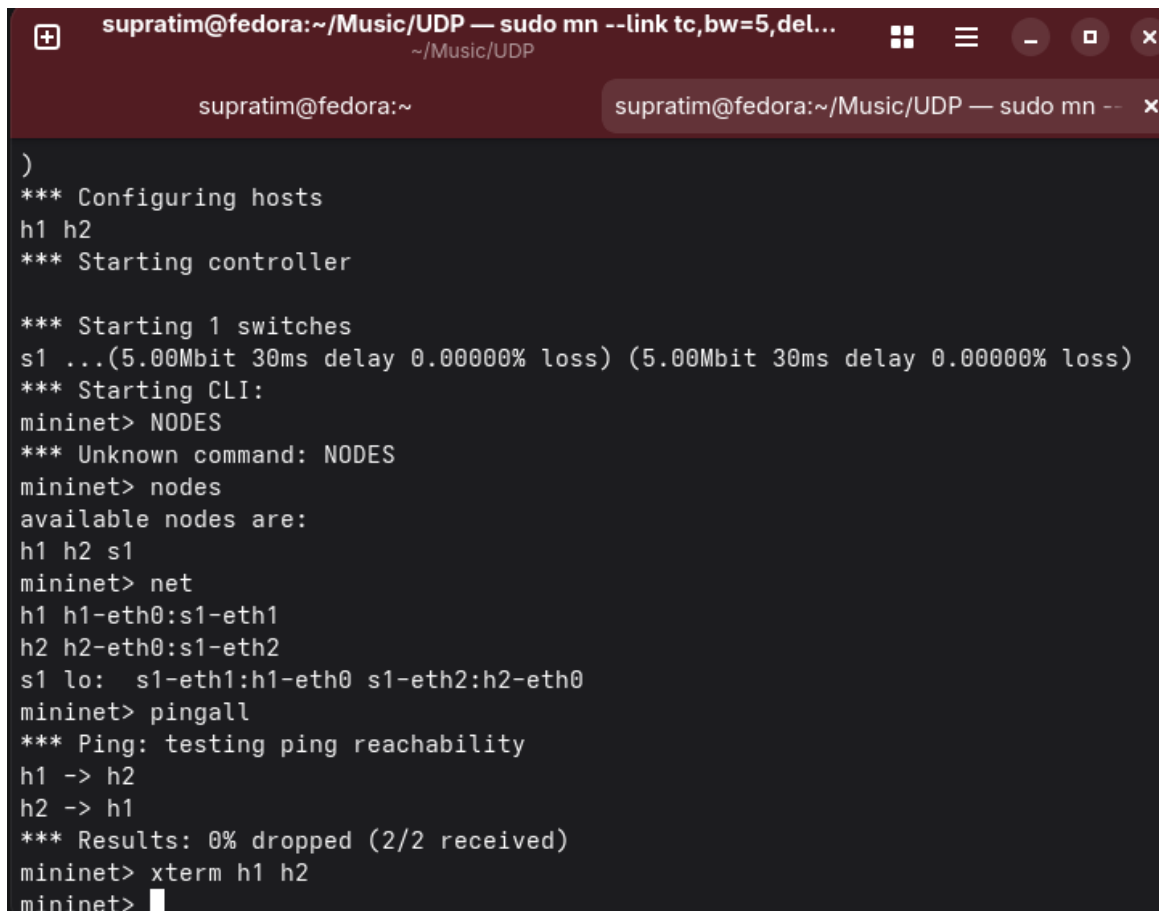
The client transmits 10 UDP messages with sequence numbers. The server acknowledges each packet. Missing ACKs trigger retransmissions after the timeout of 1.2 seconds. The server detects duplicate packets and counts unique packets.

Experimental Result Table

roll_no	name	loss_percent	timeout	total_messages	total_packets_sent	total_retransmissions	transfer_time_seconds	status
CS-BTC24-17	SUPRATIM GOGOI	0	1.2	10	10	0	1.207	SUCCESS
CS-BTC24-17	SUPRATIM GOGOI	0	1.2	10	11	1	2.4065	SUCCESS
CS-BTC24-17	SUPRATIM GOGOI	5	1.2	10	10	0	1.3268	SUCCESS
CS-BTC24-17	SUPRATIM GOGOI	5	1.2	10	15	5	7.2299	SUCCESS

Screenshots

Mininet Verification

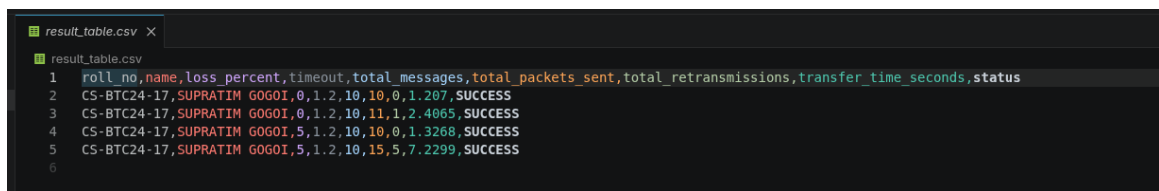


```
supratim@fedora:~/Music/UDP — sudo mn --link tc,bw=5,del...
~/Music/UDP
supratim@fedora:~
supratim@fedora:~/Music/UDP — sudo mn --

)
*** Configuring hosts
h1 h2
*** Starting controller

*** Starting 1 switches
s1 ...(5.00Mbit 30ms delay 0.000000% loss) (5.00Mbit 30ms delay 0.000000% loss)
*** Starting CLI:
mininet> NODES
*** Unknown command: NODES
mininet> nodes
available nodes are:
h1 h2 s1
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet> xterm h1 h2
mininet>
```

Program Output



```
result_table.csv
result_table.csv
1 roll_no,name,loss_percent,timeout,total_messages,total_packets_sent,total_retransmissions,transfer_time_seconds,status
2 CS-BTC24-17,SUPRATIM GOGOI,0,1.2,10,10,0,1.207,SUCCESS
3 CS-BTC24-17,SUPRATIM GOGOI,0,1.2,10,11,1,2.4065,SUCCESS
4 CS-BTC24-17,SUPRATIM GOGOI,5,1.2,10,10,0,1.3268,SUCCESS
5 CS-BTC24-17,SUPRATIM GOGOI,5,1.2,10,15,5,7.2299,SUCCESS
6
```

Questions and Answers

Why is ACK needed?

ACK confirms successful packet delivery.

Why is timeout needed?

Timeout allows retransmission if ACK is not received.

Why can duplicate messages occur?

Duplicate packets occur due to retransmissions after lost ACKs.

What happens when packet loss increases?

Transfer time and retransmissions increase.

How is this method similar to TCP?

Both use ACKs, sequence numbers and retransmissions.

How is it different from TCP?

TCP also provides connection management, flow control and congestion control.

Conclusion

Reliable UDP using Stop-and-Wait successfully delivered all messages under different packet loss conditions while demonstrating the impact of packet loss on retransmissions and transfer time.